



ST BENEDICT'S

SUBJECT	Physical Science Paper 2	DATE	5 August 2019
GRADE	11	MARKS	162
EXAMINER NAME	Mr Ntombela	MODERATOR	Mr Du Toit
CLASS		DURATION	150 minutes

**LOW
ORDER** 55%

HIGH ORDER 45%

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 10 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages (i–ii). Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae whenever necessary.
5. Start each question on a new page.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Where appropriate express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	Total
Marks	18	23	24	28	31	18	20	162

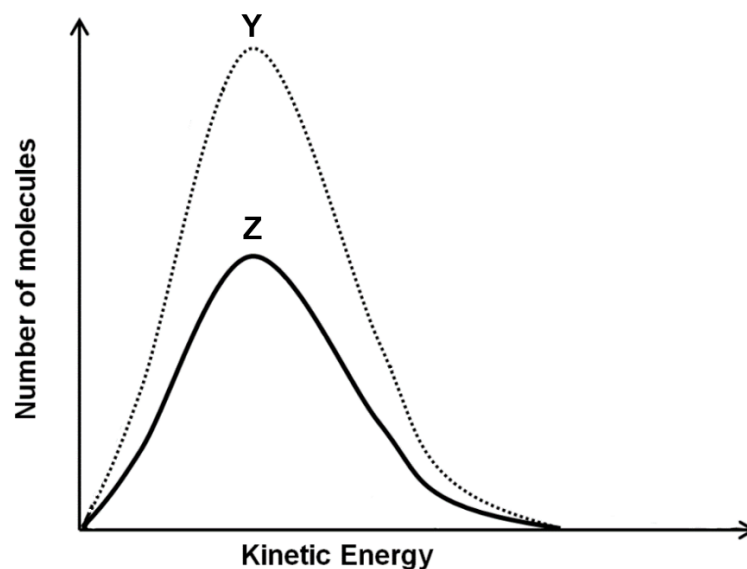
QUESTION 1- MULTIPLE CHOICE

18 MARKS

There is only one correct answer to each of the Multiple Choice questions. On the ANSWER SHEET mark the letter corresponding to the answer you think is correct with a cross.

- 1.1 Choose the correct exponent for the prefix *nano*. (2)
- A. 10^{-12}
 - B. 10^{-9}
 - C. 10^{-6}
 - D. 10^{12}
- 1.2 The correct chemical formula for potassium permanganate is (2)
- A. KMnO_4
 - B. KMnO_2
 - C. K_2MnO_4
 - D. K_2MnO_2
- 1.3 Which sample of 20 g of calcium, bromine and silver has the greatest number of moles? (2)
- A. calcium
 - B. bromine
 - C. silver
 - D. All of the above have the same number of moles in 20 g.
- 1.4 Which of the following best describes concentration? (2)
- A. mass per unit volume
 - B. moles per unit mass
 - C. grams per mole
 - D. moles per unit volume

- 1.5 The Maxwell-Boltzmann curve labelled **Z** shows the distribution of molecular energies of ozone gas (O_3).



Which of the following correctly describes the change resulting in curve **Y**? (2)

- A. An increase in concentration of ozone gas only.
- B. An increase in temperature of ozone gas only.
- C. An increase in concentration and increase in temperature of ozone gas.
- D. The addition of a catalyst.

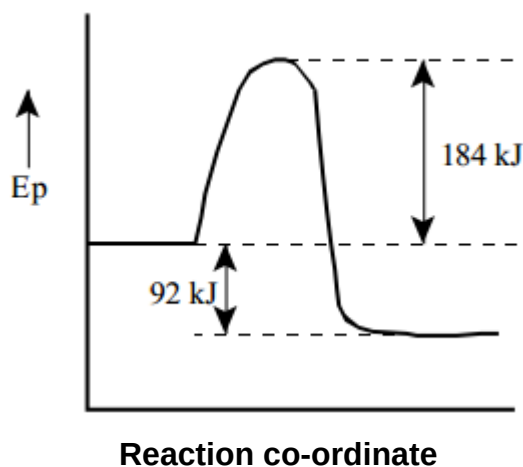
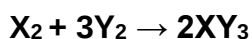
- 1.6 Anions are formed through...? (2)

- A. Oxidation of a metal
- B. Oxidation of a non-metal
- C. Reduction of a metal
- D. Reduction of a non-metal

- 1.7 During the formation of an ionic bond electrons are: (2)

- A. shared equally.
- B. shared unequally.
- C. transferred from a metal to a non-metal.
- D. transferred from a non-metal to a metal.

1.8 The following diagram illustrates the change in potential energy for the reaction



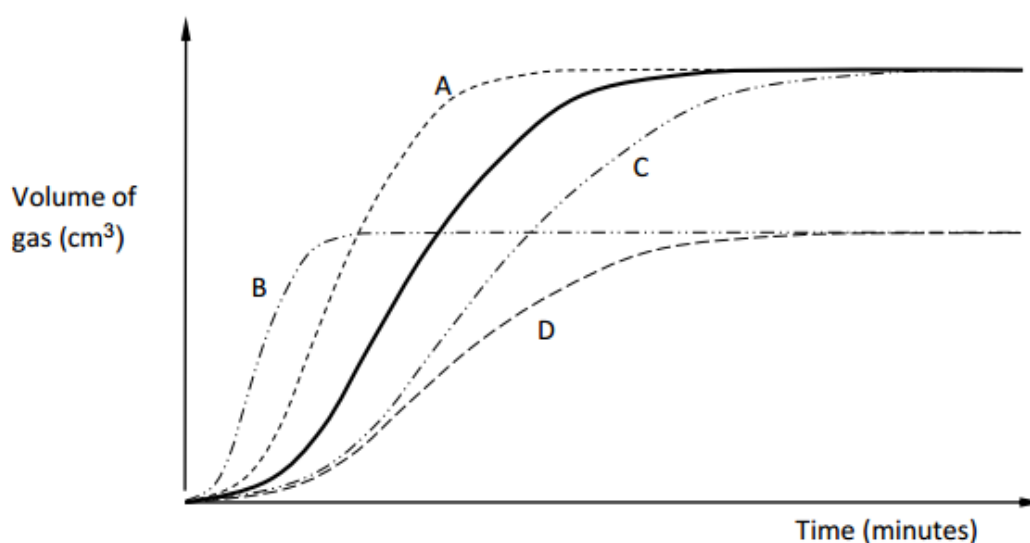
What is the heat of reaction per mole of XY_3 formed?

(2)

- A. -92 kJ
- B. -46 kJ
- C. $+92 \text{ kJ}$
- D. $+46 \text{ kJ}$

1.9 In an experiment, a 2 g lump of zinc and 1 g of powdered zinc are added separately to excess dilute sulphuric acid. The solid line in the graph below shows the volume of the gas given off when the 2 g lump is used. Which dotted line is obtained when 1 g of powdered zinc is used?

(2)



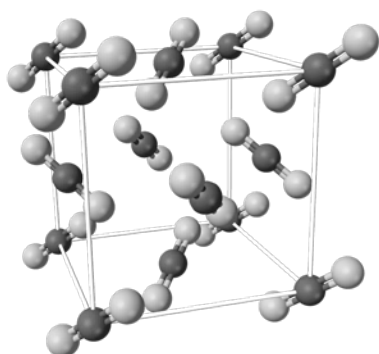
QUESTION 2

23 MARKS

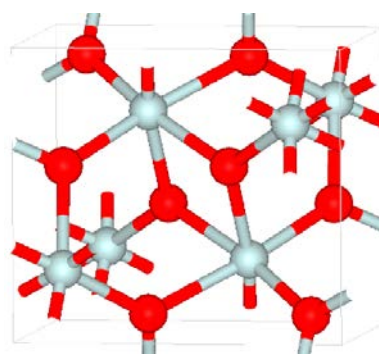
- 2.1 The table gives the melting point and boiling point of carbon dioxide (CO_2) and silicon dioxide (SiO_2).

	Carbon dioxide (CO_2)	Silicon dioxide (SiO_2)
Melting point ($^{\circ}\text{C}$)	-57	1610
Boiling point ($^{\circ}\text{C}$)	-79	2230

- 2.1.1 Define *intramolecular bond*. (2)
- 2.1.2 Give the name of the specific type of intramolecular bonds present in CO_2 molecules. (2)
- 2.1.3 In what physical state is CO_2 at room temperature (25°C)? (1)
- 2.1.4 In what physical state is SiO_2 at 2000°C ? (1)
- 2.1.5 Explain why SiO_2 has a significantly higher melting point than that of CO_2 . Refer to the following diagrams in your explanation and the relative strengths of the specific lattice structures. (5)



carbon dioxide lattice



silicon dioxide lattice

Source: Ben Mills. Crystal structure data from AMCSD.

- 2.2 A student is instructed to prepare 0,25 litre of a standard aqueous solution of oxalic acid, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$, of concentration $0,2 \text{ mol} \cdot \text{dm}^{-3}$.
- 2.2.1 Define the term *intermolecular force*. (2)
- 2.2.2 Name the intermolecular forces between the solute and the solvent in the solution. (2)
- 2.2.3 Calculate the mass of $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ required to make this standard solution. (5)
- 2.2.4 When making a standard solution, must the solute or water be added to the volumetric flask first? Explain your answer. (3)

QUESTION 3**24 MARKS**

The boiling of some Group IV and Group V hydrides are given below. The data will be needed to help answer the questions that follow.

	Group IV		Group V	
Compound	CH ₄	SiH ₄	NH ₃	PH ₃
Boiling point (°C)	-161	-112	-33	-88

- 3.1 Use bonding principles to explain fully whether a CH₄ molecule is polar or non-polar. (4)
- 3.2 NH₃ has a significantly higher boiling point than CH₄. Account for this difference with reference to the relevant intermolecular force in each of these compounds. (4)
- 3.3 Use the information in the table to determine which substance, PH₃ or CH₄, has the stronger intermolecular force. Explain your answer briefly. (2)
- 3.4 Consider pure ammonia in its liquid phase, NH₃(ℓ), which has two hydrogen bond forces between its molecules.
- 3.4.1 State two properties of the nitrogen atom that are responsible for the hydrogen bond forces. (2)
- 3.4.2 Show how two molecules of NH₃ interact in the liquid phase using Couper diagrams. Draw in the symbols δ^+ and δ^- where appropriate. (4)
- 3.4.3 Are the bonds in ammonia sigma or pi bonds? (1)
- Ammonia can form a dative covalent bond to a hydrogen ion to form the ammonium ion.
- 3.4.4 Define the term *covalent bond*. (2)
- 3.4.5 What is a dative covalent bond? (2)
- 3.4.6 Draw the ammonium ion using Lewis notation. (3)
-

QUESTION 4

28 MARKS

In order to investigate the rate of the reaction between a carbonate and an acid, calcium carbonate and an excess of $2 \text{ mol} \cdot \text{dm}^{-3}$ hydrochloric acid were placed in a reaction vessel. The balanced chemical equation for this reaction is:



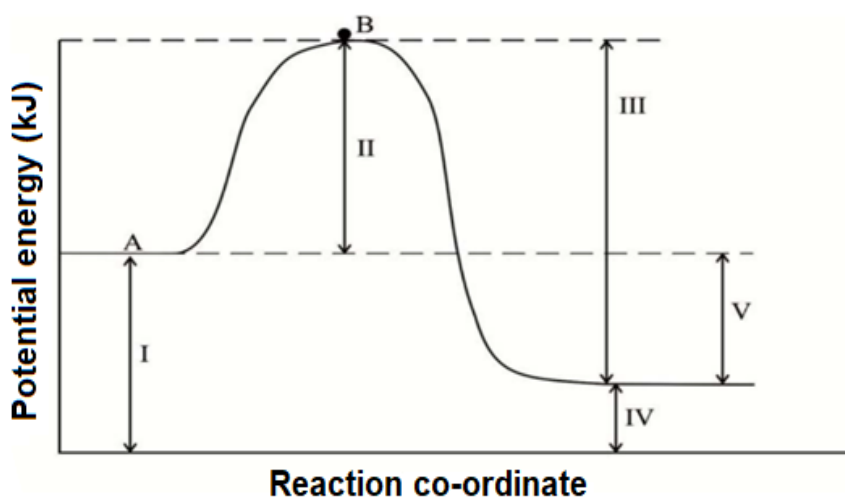
4.1 5 g of calcium carbonate was added to a $0,25 \text{ dm}^3$ solution of hydrochloric acid.

4.1.1 Calculate the number of moles of calcium carbonate that was added to the hydrochloric acid. (2)

4.1.2 Prove that calcium carbonate is the limiting reagent. (4)

4.1.3 Determine the volume of unreacted hydrochloric acid. (3)

4.2 Consider the potential energy profile for this reaction as illustrated below:



4.2.1 Define *activated complex*. (2)

4.2.2 The graph has been labelled I – V to represent the various energies that are illustrated by this energy profile. Provide the labels of these energies from I to V. (5)

4.2.3 Is the energy profile of an ENDOTHERMIC or EXOTHERMIC reaction? Explain how you came to this conclusion. (3)

4.3 The same reaction takes place in the presence of a catalyst.

4.3.1 Define *catalyst*. (2)

4.3.2 On the potential energy profile provided on your ANSWER SHEET, draw a new graph to show the effect of a catalyst on this reaction. (2)

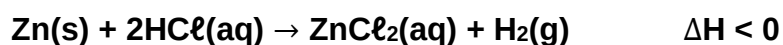
4.3.3 How would the presence of a catalyst affect the value of the energy labelled V? Explain your answer. (3)

4.3.4 A Maxwell-Boltzmann distribution for this reaction at 25°C is on your ANSWER SHEET. On the answer sheet draw a graph of the distribution for this reaction at 100°C . (2)

QUESTION 5

31 MARKS

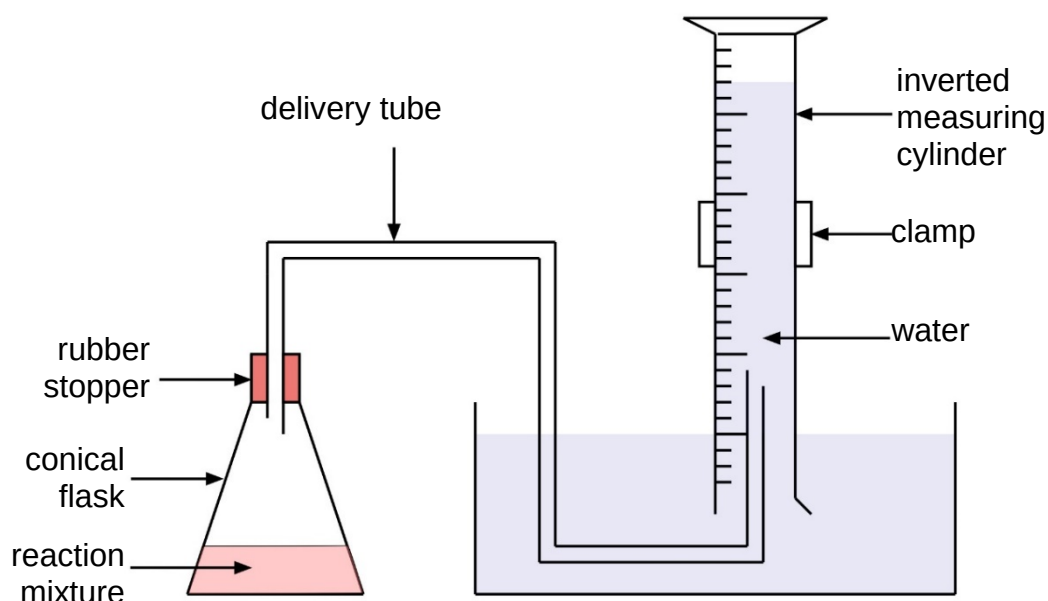
Consider the following balanced chemical equation for the reaction of zinc with hydrochloric acid.



5.1 Define *reaction rate*. (2)

5.2 Briefly explain why measuring mass over time is not a suitable technique for measuring the rate of this reaction. (2)

Scott and Riley are interested in determining how changing the concentration of HCl will affect the rate of reaction for the reaction above. Scott and Riley decide to measure the volume of H₂ gas produced over time using the following set up.



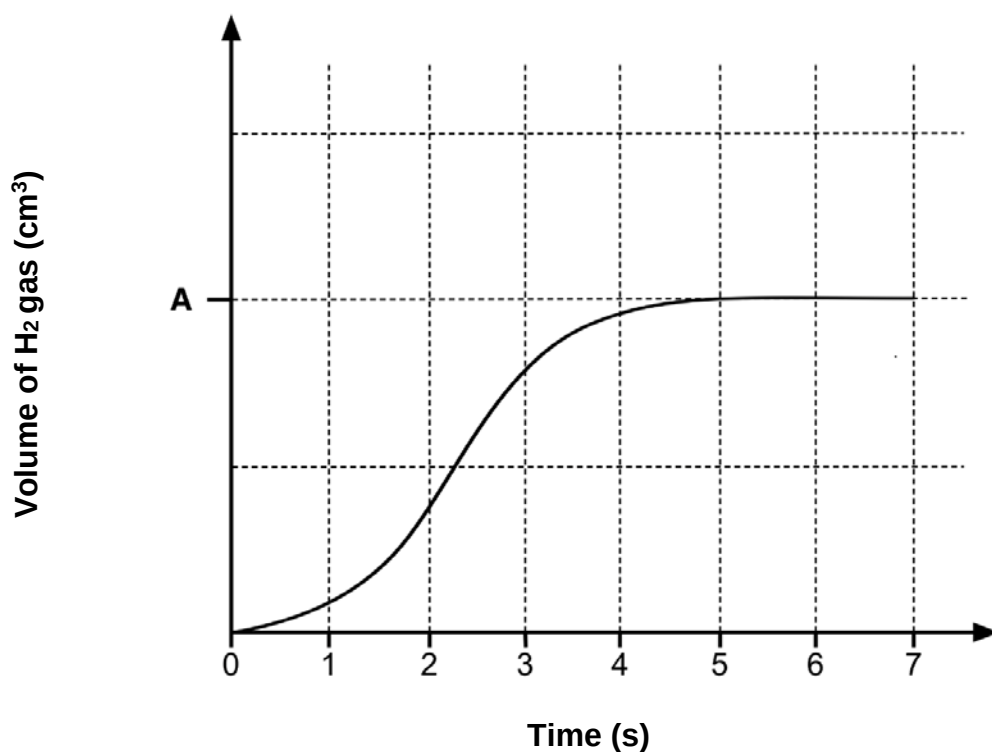
They split the workload and perform separate experiments. The following table shows the conditions for each of their experiments conducted at STP.

	Concentration of HCl (mol·dm ⁻³)	Volume of HCl (mℓ)	Mass of Zn (g)	State of Zn
SCOTT	1,0	100	0,5	Granules
RILEY	2,0	100	0,6	Powder

Note that in each experiment:

- The zinc is completely covered by the acid.
- The zinc is always the limiting reagent.

- 5.3 What is the *independent variable* for their investigation? (2)
- 5.4 State one variable that is controlled in both of their separate experiments other than the volume of HCl. (2)
- 5.5 **SCOTT** produces the following graph for his experiment by plotting the volume of H₂ gas produced over time.



According to the graph, the reaction rate initially increases (in the first two seconds) and then decreases thereafter.

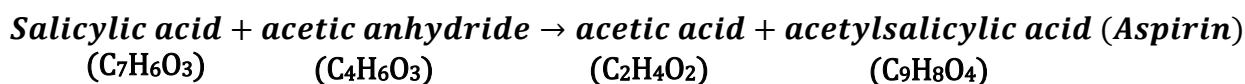
- 5.5.1 Define *heat of reaction*. (2)
- 5.5.2 Hence, describe briefly why the reaction rate would increase during the first two seconds. (2)
- 5.5.3 With reference to the hydrochloric acid, use the collision theory to explain why the reaction rate decreases after the first two seconds. (4)
- 5.5.4 Determine the volume, **A**, of H₂ produced using units **cm³**. For this calculation assume the H₂ is collected at STP. (5)
- 5.5.5 SCOTT'S teacher said that the volume for **A** should have been 200 cm³. Calculate the percentage accuracy of SCOTT'S results. (3)
- 5.5.6 The graph above is reproduced on your ANSWER SHEET. Draw a sketch of the graph you would expect from RILEY'S experiment on the graph on your ANSWER SHEET. Label this graph 'RILEY'. (3)
- 5.6 Would Scott and Riley's investigation produce reliable results? Explain your answer. (4)

QUESTION 6

18 MARKS

Acetylsalicylic Acid (also known as Aspirin) can be made in a microwave oven by reacting salicylic acid with acetic anhydride for 5 minutes. The by-product is acetic acid (also known as vinegar).

The word equation for the reaction is:



- 6.1 3,86 g of salicylic acid was reacted with an excess of acetic anhydride.
- 6.1.1 Define the term *molar mass*. (2)
- 6.1.2 Calculate the molar mass of a salicylic acid molecule (2)
- 6.1.3 Calculate the mass of aspirin that should have been produced by 3,86 g of the salicylic acid mass produced. (6)
- In reality, only 4,65 g of aspirin was produced.
- 6.1.4 Calculate the percentage yield of aspirin. (3)
- 6.2 A single aspirin tablet contains 300 mg of aspirin. Calculate the number of hydrogen atoms in the 300 mg tablet. (5)
-

QUESTION 7

20 MARKS

- 7.1 A 500 cm³ aqueous solution of sodium sulphate with a concentration of 1,5 mol·dm⁻³ is prepared in the lab. The solution is then diluted to a concentration of 0,25 mol·dm⁻³.
- 7.1.1 Define the term *solvent*. (2)
- 7.1.2 Identify the solvent in the above solution. (1)
- 7.1.3 Calculate how much water (in dm³) was added to the solution to dilute it to a concentration of 0,25 mol·dm⁻³. (5)
- 7.2 Two students prepare 200 cm³ of a standard aqueous solution of magnesium nitrate (Mg(NO₃)₂ (aq)) of concentration 0,15 mol·dm⁻³.
- 7.2.1 Write down a balanced chemical equation for the dissociation of Mg(NO₃)₂ in water. Phase indicators (state symbols) must be shown. (4)
- 7.2.2 Calculate the number of moles of nitrate ions (NO₃⁻) in the solution. (3)
- The students now dissolves **solid** sodium nitrate (NaNO₃) in the magnesium nitrate solution. The total concentration of nitrate ions (NO₃⁻) in the **new** solution is 0,50 mol·dm⁻³. (Assume that there is no change in the volume of the solution).
- 7.2.3 Calculate the mass of sodium nitrate dissolved. (5)
-

END OF EXAMINATION



ST BENEDICT'S

PHYSICAL SCIENCE
Grade 11 Exam
PAPER 2 (5 August 2019)

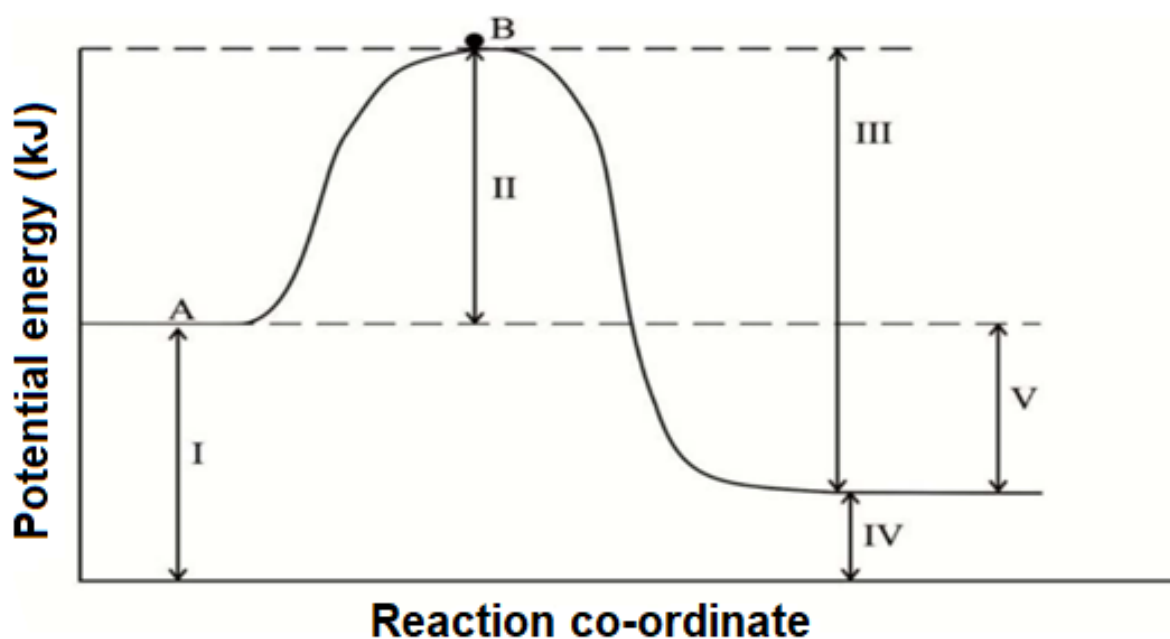
ANSWER SHEET

NAME: _____

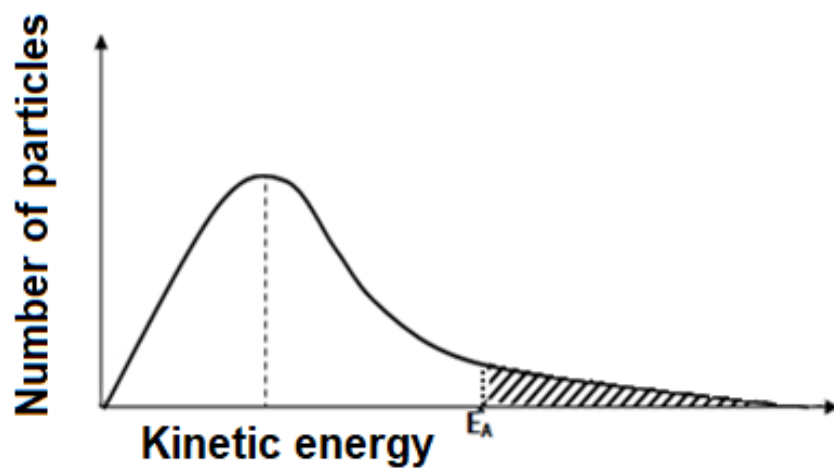
QUESTION 1

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D

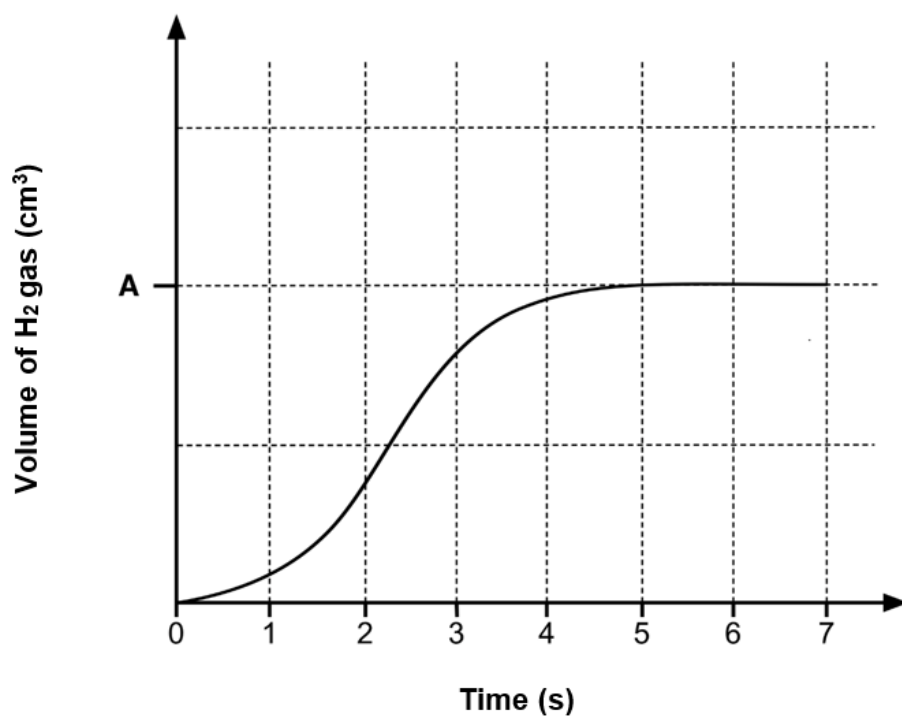
QUESTION 4.3.2



QUESTION 4.3.4



QUESTION 5.5.6



EXAMINATION DATA SHEET FOR THE PHYSICAL SCIENCES (CHEMISTRY)

TABLE 1 PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Magnitude of charge on electron	e	$1,6 \times 10^{-19} \text{ C}$
Mass of an electron	m_e	$9,1 \times 10^{-31} \text{ kg}$
Standard pressure	p^θ	$1,01 \times 10^5 \text{ Pa}$
Molar gas volume at STP	V_m	$22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$
Standard temperature	T^θ	273 K
Avogadro's constant	N_A	$6,02 \times 10^{23} \text{ mol}^{-1}$
Faraday's constant	F	$96\,500 \text{ C} \cdot \text{mol}^{-1}$

TABLE 2 CHEMISTRY FORMULAE

$n = \frac{m}{M}$	$n = \frac{N}{N_A}$	$n = \frac{V}{V_m}$
$c = \frac{n}{V}$ OR $c = \frac{m}{MV}$	$K_w = [H_3O^+] \cdot [OH^-] = 1 \times 10^{-14}$ at 25 °C (298 K)	
$q = It$ $q = nF$	$E_{cell}^\theta = E_{cathode}^\theta - E_{anode}^\theta$ $E_{cell}^\theta = E_{oxidising\ agent}^\theta - E_{reducing\ agent}^\theta$	

TABLE 3 PERIODIC TABLE

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Atomic number (Z)																	
	Electronegativity																	
1	1 2,1 H																	2 He
2	3 1,0 Li	4 1,5 Be											5 2,0 B	6 2,5 C	7 3,0 N	8 3,5 O	9 4,0 F	10 Ne
3	11 0,9 Na	12 1,2 Mg											13 1,5 Al	14 1,8 Si	15 2,1 P	16 2,5 S	17 3,0 Cl	18 Ar
4	19 0,8 K	20 1,0 Ca	21 1,3 Sc	22 1,5 Ti	23 1,6 V	24 1,6 Cr	25 1,5 Mn	26 1,8 Fe	27 1,8 Co	28 1,8 Ni	29 1,9 Cu	30 1,6 Zn	31 1,6 Ga	32 1,8 Ge	33 2,0 As	34 2,4 Se	35 2,8 Br	36 Kr
5	37 0,8 Rb	38 1,0 Sr	39 1,2 Y	40 1,4 Zr	41 1,6 Nb	42 1,8 Mo	43 1,9 Tc	44 2,2 Ru	45 2,2 Rh	46 2,2 Pd	47 1,9 Ag	48 1,7 Cd	49 1,7 In	50 1,8 Sn	51 1,9 Sb	52 2,1 Te	53 2,5 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra																

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lw